

Voice to Vision: A Sociotechnical System for Transparent Civic Decision-Making

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Abstract

Communities frequently report sending feedback “into a void” during community engagement processes like neighborhood planning, creating a critical disconnect between public input and decision-making. Voice to Vision addresses this gap with a sociotechnical system that comprises three integrated components: a flexible data architecture linking community input to planning outputs, a sense-making interface for planners to analyze and synthesize feedback, and a community-facing platform that makes the entire engagement process transparent. By creating a shared information space between stakeholders, our system demonstrates how structured data and specialized interfaces can foster cooperation across stakeholder groups, while addressing tensions in accessibility and trust formation. Our CSCW demonstration will showcase this system’s ability to transform opaque civic decision-making processes into collaborative exchanges, inviting feedback on its potential applications beyond urban planning.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; *Visualization systems and tools*; **Collaborative and social computing systems and tools**; • **Applied computing** → **Computing in government**.

Keywords

Democracy, Civic Tech, Community Engagement, Transparency in Government, Data Visualization

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1 Introduction

Trust and legitimacy in civic processes are eroding as communities increasingly report sending feedback “into a void” without understanding how their input influences outcomes [4]. This disconnect is particularly acute in urban planning, where complex decisions affect thousands of residents, yet the process of transforming community feedback into concrete planning outputs remains opaque [7]. Voice to Vision addresses this gap through an integrated sociotechnical system that supports planners and community members. The system comprises three components: a flexible data architecture that connects community input to planning outputs; a sensemaking interface for planners to organize and synthesize feedback; and a community-facing platform that makes engagement processes transparent. This approach builds upon prior work in civic technology (e.g. [1, 2]) but uniquely focuses on creating a comprehensive data infrastructure that allows stakeholders to trace how community voices influence planning decisions.

Our approach aligns with multiple CSCW interests, including visualization of collaborative workflows, support for multi-stakeholder and community-informed decision-making, and fostering cooperation across power differentials [3, 6]. By promoting shared understanding between planners and communities, Voice to Vision demonstrates how collaborative sensemaking technologies can strengthen democratic participation while addressing critical tensions in accessibility, actionability, and trust formation in civic contexts. We propose demonstrating this system at CSCW to gather feedback on its potential applications beyond urban planning to other cooperative contexts.

2 Voice to Vision System

Voice to Vision is a sociotechnical system designed to bridge community input and planning outputs through a structured data architecture and specialized interfaces for both community members and planners. The system comprises three integrated components: a flexible data structure, a sensemaking interface, and a community-facing platform.

Data Structure and Technical Architecture: At the core of Voice to Vision lies a structured yet adaptable data model organized into interconnected collections (Table 1). The data structure employs a flexible schema model that accommodates the inherently unstructured nature of community input while maintaining consistent relationships between collections. This design balances consistency through defined data fields with flexibility through optional fields and JSON-compatible inputs. The data collections also enable voices to be linked directly to planning outputs, creating explicit connections between what community members say and how their feedback influences decisions. The system's technical implementation uses React for frontend interfaces, Flask as the backend web framework, MongoDB as the database, and Docker containerization deployed through Amazon Web Services.

Sensemaking Interface: The sensemaking interface helps planners organize, analyze, and synthesize community feedback into actionable outputs. Voice cards with editable metadata enable planners to categorize and connect feedback to planning outputs, while output creation tools link community input directly to goals and strategies. Interactive visualizations (Figure 1), including maps and cluster views, reveal spatial and thematic patterns across voices. The sensemaking interface supports systematic data analysis workflows by enabling planners to find connections across diverse inputs and structure outputs in formats that can be accessibly communicated back to the community. Through iterative design with professional planners, we optimized the interface to support both exploration and synthesis.

Community-Facing Platform: The community-facing platform makes engagement processes transparent and accessible to the public. Voice exploration through read-only cards (Figure 2 left) displays community input alongside metadata about engagement events and connections to outcomes. Output cards (Figure 2 right) show how community feedback informed specific goals and strategies. The interactive visualizations and data exploration capabilities in the sensemaking interface are also present in the community-facing platform.

The platform addresses common frustrations in community engagement by making visible how input influenced decisions, providing multiple entry points for exploring feedback, and creating transparency around the planning process. Through its integrated components, Voice to Vision creates a shared information space where both planners and community members can develop a common understanding of neighborhood needs and priorities.

While our initial implementation focuses on urban planning, the system's architecture is deliberately designed to support a wide range of cooperative contexts where multi-stakeholder input must be systematically analyzed and incorporated into decisions. The flexible data model can be adapted for contexts including participatory budgeting, academic conference organization, or any scenario

where diverse voices must be heard; synthesis is required; and transparent decision-making is valued. This adaptability makes Voice to Vision relevant to CSCW researchers studying various forms of collaborative sensemaking across organizational boundaries and power differentials.

3 Discussion and Future Work

Our initial evaluation of Voice to Vision involved user studies with both urban planners (N=7) and community members (N=17). Through semi-structured interviews, we gathered preliminary insights about the system's potential to transform community engagement practices. While this initial assessment is limited in scale, it revealed promising directions for both system refinement and broader application across cooperative contexts.

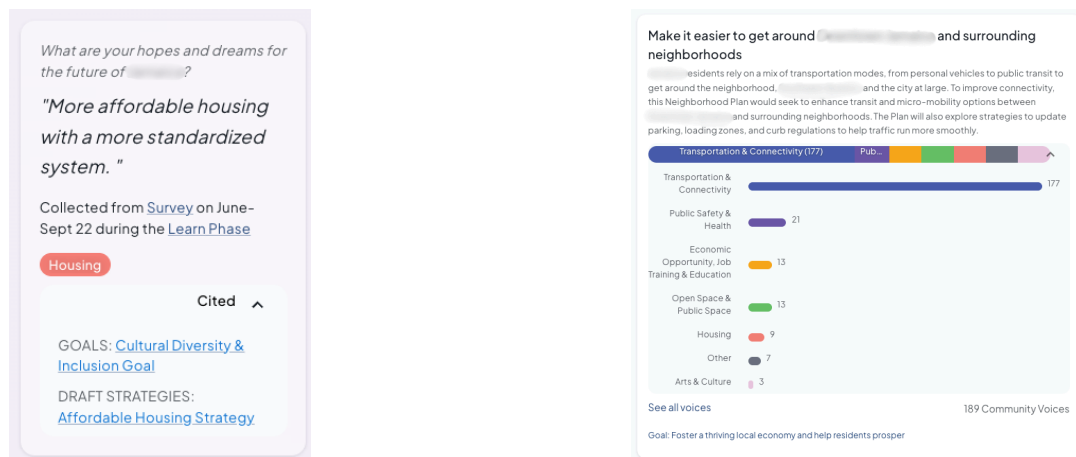
Early feedback suggests the system effectively connects community input to planning outputs by making relationships explicit and transparent. Planners valued the sensemaking capabilities that helped reveal patterns across diverse inputs, while community members appreciated seeing how feedback influenced decisions. A recurring theme across stakeholder groups was the tension between providing digestible information summaries versus supporting deeper exploration [5], which suggests future work on adaptive interfaces that respond to varying information needs. As we continue development, we are particularly interested in CSCW community feedback on how our approach might extend to other cooperative contexts where stakeholder power differentials exist, such as educational settings or corporate-community partnerships. The demo session offers an opportunity to explore how Voice to Vision might evolve from a planning-specific tool into a flexible platform supporting diverse sensemaking needs while preserving the core value of connecting voices to outcomes through structured yet flexible data architectures.

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Table 1: Key data collections in the Voice to Vision system architecture.

Collection	Description
Project	Stores contextual information about the community engagement process, including phase details.
Events	Captures diverse feedback-gathering mechanisms ranging from surveys to community workshops and tabling events, including descriptive information for each engagement activity.
Voices	Records community input with associated metadata connecting voices to events, topics, and outputs, forming the central repository of community feedback that drives subsequent analysis.
Topics	Maintains information about overarching themes used to organize and make sense of community input. These topics may be predefined or emerge organically through the sensemaking process.
Outputs	Stores the synthesized products of the engagement process, categorized as insights (key findings), goals (high-level visions), and recommendations (specific strategies). Beyond storing descriptive information and voice citations, this collection captures connections between different outputs.

**Figure 1: The map visualization (left) and cluster visualization (right) in the sensemaking interface.****Figure 2: Read-only voice (left) and output (right) cards in community-facing platform.**